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DC Voltage Prediction based on CNN with Physics Information Embedding for MVDC Distribution Systems

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***Abstract*—Conventional neural networks (NNs), though efficient in rapid dc voltage calculations for medium-voltage direct current (MVDC) distribution systems with diverse converter control schemes, face accuracy challenges with untrained system parameter distributions and topology structures. This paper proposes a physics-embedded convolutional NN (PECNN) to address this issue. The PECNN is enhanced with two additional layers, including a multi-channel combination layer and a physics operation layer. The former channel combination layer strengthens the model feature extraction capability by converting all input channels constituted by initial data matrix into dc voltage, current and line conductance matrix channels. The latter physics operation layer reformulates the combined input channels by physical connections in MVDC systems. The new layers enhance the prediction accuracy and allow generalization of the model. Three MVDC distribution networks with different dc voltage levels and network layouts are used to verify the superiority of proposed PECNN compared to other NNs in different topologies.**

***Index Terms*—Medium-voltage dc (MVDC) system, dc voltage estimation, neural network (NN), convolutional NN (CNN).**

I. Introduction

Medium-voltage dc (MVDC) power systems have the potential to outperform equivalent ac counterparts in many applications, especially in distribution grids. MVDC distribution systems offer a range of network configurations, including radial, ring, and meshed structures, enhanced by the integration of multiple ac/dc and dc/dc converters [1], [2]. For improved security and planning in MVDC networks, it is vital to conduct power flow analysis for different operation scenarios as line disconnection and converter outage, with a key focus on accurately deriving terminal dc voltage [3].

Many numerical methods can be employed for dc voltage calculation in MT-MVDC networks. Methods represented by the Newton-Raphson (NR) are effective for an MVDC distribution system that utilize various converters with diverse control schemes [4]. However, numerical methods come with certain disadvantages such as convergence for ill-conditioned systems, and computational complexity for large systems [4], [5]. With advances in computer processing capabilities, the use of neural networks (NNs) is becoming more prevalent. Once appropriately trained, NNs can predict results in a single forward propagation, outpacing numerical methods [6]. An MVDC distribution system often experiences shifts in steady-state operation points due to the presence of time-variable loads, distributed generation (DG), and energy storage systems (ESSs) with power sharing achieved through connected converters utilizing various control schemes [7]. This underscores the necessity for employed NNs to possess excellent generalization capabilities and adapt to varying operational scenarios and ensure stable system performance.

Multilayer perceptron (MLP) is the widely used artificial neural networks (ANNs) in power flow analysis [8]. MLP can also support power flow analysis even in the absence of comprehensive initial system data [9]. It is also employed in optimal power flow to enhance the efficiency of traditional methods [10] or to directly produce optimal solutions [11]–[13]. Although MLP is effective in fixed topology structures, they encounter challenges when the system topologies shift [14]. Compared to conventional MLP, convolutional NN (CNN) inherently has the capability to handle input with variable sizes due to spatial hierarchies and local connectivity patterns, which allow adoption of different spatial dimensions without requiring a fixed-size input vector [15].

CNN has been extensively employed in power flow studies under uncertain contingency scenarios [16], demonstrating efficiencies up to 100 times [17] and 350 times [18], [19] that of traditional numerical methods for optimal power flow solutions. In addition, the Latin-hypercube sampling method has been incorporated into a CNN to enhance training efficiency

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in probabilistic power flow analyses [20]. The integration of a CNN with long short-term memory (LSTM) networks facilitates the inclusion of weather effects in power flow determinations [21]. Nevertheless, the traditional approach to CNN channel construction fails to fully capture the complex relationships between node and line parameters such as node voltage, power and line impedance that constitute the inherent physical connections in power systems. This lack of representation leads to diminished performance when dealing with unseen data, as it overlooks the interactions between these key elements that fundamentally drive the electrical network behaviors [22].

This paper proposes a novel physics-embedded CNN (PECNN), specifically designed for terminal dc voltage estimation in MVDC distribution systems with different converter control schemes. In this model, the input channels are reformulated to account for the relationships between voltage, power, current at MVDC terminals and line parameters. The voltage and current channels mutually transform through the line conductance channel, reflecting the actual voltage and current interactions. Moreover, a more accurate representation of the physical connection is captured after performing multiple reformulation calculations. This operation enhances both the prediction accuracy and generalization capability, as it allows the network to comprehend the complex interaction between voltage, current, and line conductance in an MVDCs system.

II. Numerical Computation Method

Different control schemes can be adopted in dc terminal-linked converters in an MVDC distribution system, including *i*) dc bus voltage control, *ii*) dc power or current control, and *iii*) power/voltage (P/V) or current/voltage (I/V) droop control [3]. Five MVDC bus types can be defined to compute the terminal voltage based on a NR method in an MVDC system with n buses. They are one MVDC voltage bus (slack bus), m dc power buses, l dc current buses, k P/V droop-controlled buses, $(n-m-l-k-1)$ I/V droop-controlled buses.

Eq. (1) summarizes the PF equations for dc voltage, dc power, dc current, P/V droop-controlled and I/V droop-controlled buses, respectively.

$$\begin{cases} V_{dcrefm,i} - V_{dcm,i} = 0, \qquad (i=1) \\ P_{dcrefm,i} - P_{dcm,i} = 0, \qquad (i=2,...,m+1) \\ I_{dcrefm,i} - I_{dcm,i} = 0, \qquad (i=m+2,...,l+m+1) \\ (P_{dcrefm,i} - P_{dcm,i}) + K^{PV}_{droop,i}(V_{dcrefm,i} - V_{dcm,i}) = 0 \\ (i=l+m+2,...,k+l+m+1) \\ (I_{dcrefm,i} - I_{dcm,i}) + K^{IV}_{droop,i}(V_{dcrefm,i} - V_{dcm,i}) = 0 \\ (i=k+l+m+2,...,n) \end{cases}, \tag{1}$$

where V_{dcrefm} is usually set as rated voltage V^{rated}_{dcm}, $K^{PV}_{droop,i}$ and $K^{IV}_{droop,i}$ are the droop constants for converters with P/V and I/V droop, respectively:

$$K^{PV}_{droop,i} = (V^{rated}_{dcm}\delta_{droop})^{-1}P^{rated}_{dc}, \tag{2}$$

$$K^{IV}_{droop,i} = (V^{rated}_{dcm} \cdot V^{rated}_{dcm}\delta_{droop})^{-1}P^{rated}_{dc}, \tag{3}$$

and δ_{droop} is the maximum allowable dc voltage deviation ratio [23]. Moreover, the dc power and current injected to any MVDC buses is expressed as:

$$\begin{cases} P_{dcm,i} = V_{dcm,i}\sum_{j=1}^{n} G_{dcm,ij}V_{dcm,j} \\ I_{dcm,i} = \sum_{j=1}^{n} G_{dcm,ij}V_{dcm,j} \end{cases} \tag{4}$$

where $G_{dcm,ij}$ is the line conductance in an MVDC system. The dc voltage in all MVDC terminals can be derived by executing NR algorithm [24].

Although the conventional numerical approach can derive the terminal voltage of MVDC distribution systems accurately, it requires repetitive computations resulting in a significant computational expense. Therefore, it is not well-suited for MVDC systems with many time-variable loads, DGs and ESSs that require rapid voltage computations.

III. Proposed PECNN for MVDC Distribution Networks

A. General Network Description

Fig. 1 shows the proposed PECNN which comprises *i*) one multi-channel combination layer, *ii*) one physics operation layer, *iii*) four sets of six encoder blocks with two convolutional layers. The training data is generated from real-time models built in RSCAD. These convolutional layers specifically designed to extract feature maps from inputs (F) using specified convolution kernel (K). Moreover, an activation function $\sigma(\cdot)$ offers nonlinearity to the PECNN. The first convolution layer C_f connected to the input map follows the forward propagation rule as [15]:

$$C_f = \sigma\left(F * K_{1,f} + B_f\right) \tag{5}$$

where f refers to the fth feature map, $*$ denotes to convolution operator, B is bias. The FC layer is used to generate the network output O_{PECNN} by flattening the output from encoder blocks as:

$$O_{PECNN} = \sigma(W_{fc}h + B_{fc}), \tag{6}$$

where W_{fc} and B_{fc} are the weight and bias in hidden layers h, respectively [25].

The input configuration of the network omits specifying the dimensions, such as height and width, of the input data, ensuring adaptability to various sizes. Moreover, the proposed PECNN excludes a pooling layer to avoid the potential loss of critical information, which is essential for maintaining the physical relationships among different input channels.

B. Physics-embedded Channel Reformulation

The PECNN offers improved prediction accuracy by the integration of restructured input channels. Based on MVDC PF equations (1), six input channels are designed including

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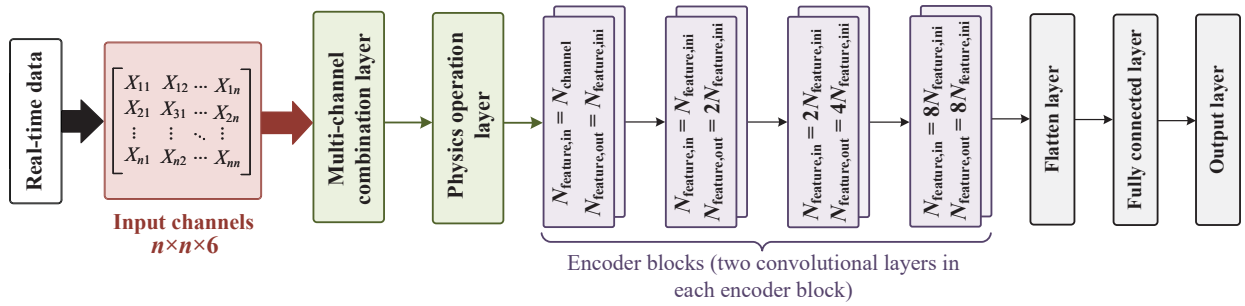

Fig. 1. Proposed PECNN structure for MVDC distribution systems.

voltage, power, current, P/V droop, I/V droop and line conductance channels as:

$$\begin{cases} \boldsymbol{CH_V} = \text{diag}(V_{dcrefm,1}, V_{dcrefm,1}, ..., V_{dcrefm,1})^{n\times n} \\ \boldsymbol{CH_P} = \text{diag}(0, P_{dcrefm,2}, ..., P_{dcrefm,m+1}, ..., 0)^{n\times n} \\ \boldsymbol{CH_I} = \text{diag}(0, ..., I_{dcrefm,m+2}, .., I_{dcrefm,l+m+1}, ..., 0)^{n\times n} \\ \boldsymbol{CH_{P/V}} = \text{diag}(0, ..., P_{dcrefm,l+m+2} + K^{PV}_{droop,l+m+2} V_{dcrefm,l+m+2}, \\ .., P_{dcrefm,k+l+m+1} + K^{PV}_{droop,k+l+m+1} V_{dcrefm,k+l+m+1}, ..., 0)^{n\times n} \\ \boldsymbol{CH_{I/V}} = \text{diag}(0, ..., I_{dcrefm,k+l+m+2} + K^{IV}_{droop,k+l+m+2} \times \\ V_{dcrefm,k+l+m+2}, .., I_{dcrefm,n} + K^{IV}_{droop,n} V_{dcrefm,n})^{n\times n} \\ \boldsymbol{CH_G} = \boldsymbol{G_{dcm}} \end{cases} \tag{7}$$

However, the potential for information extraction could be constrained when dealing with a channel that contains less effective information (fewer number of buses for a specific bus type). Fewer input channels are required for simplifying computations in forward/backward information propagations. The matrices in (7) related to power, P/V droop, and I/V droop channels are transformed into dc current matrices by:

$$\begin{cases} \boldsymbol{CH_P} \xrightarrow[i=2,...,m+1]{P_{dcrefm,i}/V^{rated}_{dcm}} \boldsymbol{CH^P_I} \\ \boldsymbol{CH_{P/V}} \xrightarrow[i=l+m+2,...,k+l+m+1]{P_{dcrefm,i}/V_{dcrefm,i}} \boldsymbol{CH^{P/V}_I} \\ \boldsymbol{CH_{I/V}} \xrightarrow[i=k+l+m+2,...,n]{\text{Only keep } I_{dcrefm,i}} \boldsymbol{CH^{I/V}_I} \end{cases} \tag{8}$$

They are then merged with the current matrix $\boldsymbol{CH_I}$ in (7) to create an updated current matrix as (9), hence the six input channels are now converted into three channels (voltage, current, conductance).

$$\boldsymbol{CH_{I,\text{total}}} = \boldsymbol{CH^P_I} + \boldsymbol{CH_I} + \boldsymbol{CH^{P/V}_I} + \boldsymbol{CH^{I/V}_I} \tag{9}$$

However, further conversion is required to enhance the model performance to diverse input network structures. Terminal current I_{dcm} has a linear correlation with the terminal voltage V_{dcm} (4), thus (10) can be established for any input power or current vectors derived from (4) and (8) by splitting self-conductance and mutual conductance.

$$\mathcal{I}^{\text{merge}}_{dcm,i} = \sum_{j=1}^{n} G_{dcm,ij} \cdot V_{dcm,j}, \tag{10}$$

where $\mathcal{I}^{\text{merge}}_{dcm,i}$ includes accurate and estimated dc current values. Eq. (10) can also be written as (11) to reflect how one node voltage is influenced by its neighbouring node voltage values.

$$V_{dcm,i} = \frac{\mathcal{I}^{\text{merge}}_{dcm,i} - \sum_{j\neq i}^{n} G_{dcm,ij} \cdot V_{dcm,j}}{G_{dcm,ii}} \tag{11}$$

The input voltage and current matrices are further reconstructed by (12) and (13), respectively, using matrix $G_{dcm,ij}$.

$$\boldsymbol{\mathcal{I}^{\text{merge,new}}_{dcm}} \leftarrow \boldsymbol{G_{dcm}} \cdot \text{diag}(\boldsymbol{V_{dcm}}) \tag{12}$$

$$\boldsymbol{V^{\text{new}}_{dcm}} \leftarrow \frac{\text{diag}(\boldsymbol{\mathcal{I}^{\text{merge}}_{dcm}}) - \text{nondiag}(\boldsymbol{G_{dcm}}) \cdot \text{diag}(\boldsymbol{V_{dcm}})}{\text{diag}(\boldsymbol{G_{dcm}})} \tag{13}$$

The reconstructed voltage and current matrices consist of N dc voltage and current components:

$$\boldsymbol{\mathcal{X}^{\text{new}}_{dcm}} = \begin{bmatrix} \mathcal{X}_{dcm,11} & \mathcal{X}_{dcm,12} & \cdots & \mathcal{X}_{dcm,1N} \\ \mathcal{X}_{dcm,21} & \mathcal{X}_{dcm,21} & \cdots & \mathcal{X}_{dcm,2N} \\ \vdots & \vdots & \ddots & \vdots \\ \underbrace{\mathcal{X}_{dcm,n1}}_{\text{component 1}} & \underbrace{\mathcal{X}_{dcm,n2}}_{\text{component 2}} & \cdots & \underbrace{\mathcal{X}_{dcm,nN}}_{\text{component N}} \end{bmatrix}, \tag{14}$$

where $\mathcal{X}$ refers to V or $\mathcal{I}$ in (12) and (13), respectively. The sum of all components in each row of (14) is the actual dc voltage or current values as:

$$\begin{cases} \sum_{row} V^{\text{new}}_{dcm,i} = \sum_{N=1}^{N_{\text{column}}} V^{\text{new}}_{dcm,iN} \\ \sum_{row} \mathcal{I}^{\text{new}}_{dcm,i} = \sum_{N=1}^{N_{\text{column}}} \mathcal{I}^{\text{new}}_{dcm,iN} \end{cases} \tag{15}$$

In voltage channel $\boldsymbol{CH_V}$, the 2nd to n^{th} diagonal voltage elements are substituted by corresponding elements in $V^{\text{new}}_{dcm,i}$.

$$CH_{Vi} = \sum_{row} V^{\text{new}}_{dcm,i} \quad (i = 2, 3, ..., n). \tag{16}$$

In current channel $\boldsymbol{CH_{I,\text{total}}}$, dc current elements in addition to the elements related to the current buses should be all updated based on the following rule:

- Voltage bus-related element:

$$CH_{Ii,\text{total}} = \sum_{row} \mathcal{I}^{\text{merge,new}}_{dcm,i} \tag{17}$$

- Power bus-related elements:

$$CH_{Ii,\text{total}} = P_{dcrefm,i} / \sum_{row} V^{\text{new}}_{dcm,i} \tag{18}$$

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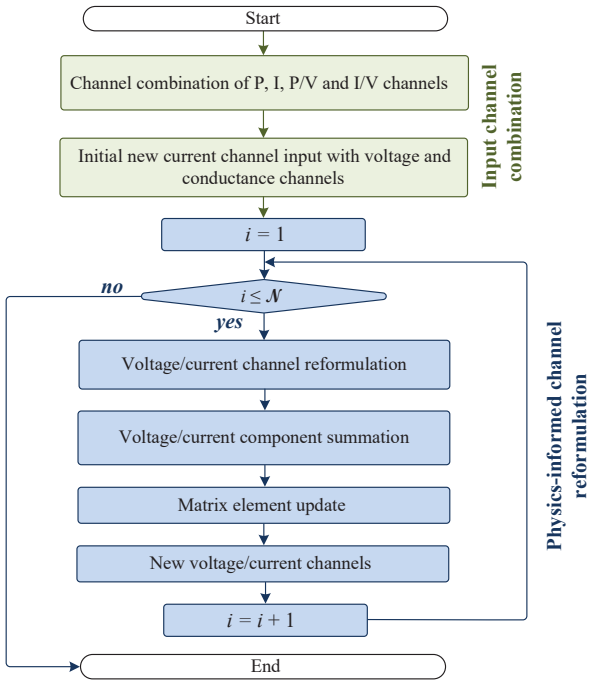

Fig. 2. Physics information embedding process in the proposed PI-FCN for MVDC distribution systems.

- P/V bus-related elements:

$$CH_{Ii,\text{total}} = \frac{P_{dcrefm,i} + K^{PV}_{droop,i}(V_{dcrefm,i} - \sum_{row} V^{\text{new}}_{dcm,i})}{\sum_{row} V^{\text{new}}_{dcm,i}} \tag{19}$$

- I/V bus-related elements:

$$CH_{Ii,\text{total}} = P_{dcrefm,i} + K^{IV}_{droop,i}(V_{dcrefm,i} - \sum_{row} V^{\text{new}}_{dcm,i}) \tag{20}$$

The prediction accuracy of the model can be improved due to the embedded physics principle. Utilizing a single channel conversion calculation by (12) and (13) yields limited physics information. By employing multiple conversion calculations, one can obtain adequate physics insights. The complete flowchart illustrating the physics information embedding process is shown in Fig. 2.

IV. Case Study

The proposed PECNN along with other NN models are developed using PyTorch. Two cases are considered to validate the superior performance of the proposed PECNN in dc voltage estimation for MVDC distribution networks compared to conventional CNN and MLP. All evaluations were conducted on a PC equipped with an Intel(R) Core(TM) i7-13700K CPU@3.4GHz, 64GB RAM, and a NVIDIA GeForce RTX 4090 GPU. The two cases include: 1) fixed network topology that five designed MVDC networks as shown in Fig. 3(a)-(e) are trained using random sampling of reference values for grid-connected, DG-connected, ESS-connected, and load-connected converters; 2) varying network topology that random line disconnections are considered. The training data for the designed MVDC systems are obtained from real-time simulation in RTDS (Fig. 4).

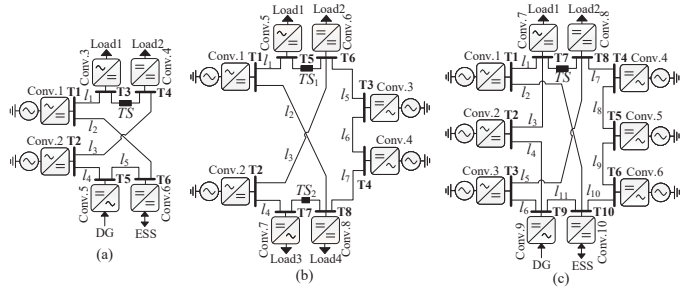

Fig. 3. MVDC network structures: (a) 6T-MVDC system with load/DG/ESS, (b) 8T-MVDC system with load, and (c) 9T-MVDC system with load/DG/ESS.

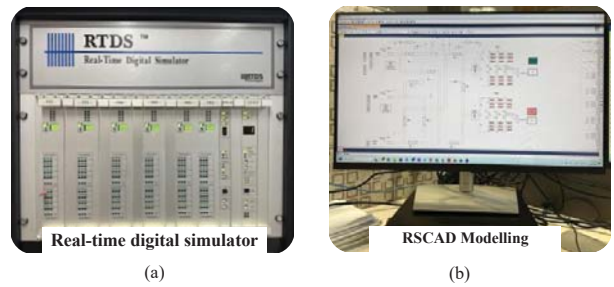

Fig. 4. Real-time data acquisition: (a) real-time digital simulator, and (b) modeling in RSCAD.

The initial parameters of converters and lines for the five networks are listed in Table I. To obtain enough training data, random sampling for the two cases follows:

- Voltage reference values (p.u.) of all converters are uniformly varied from 0.9 to 1.1.
- For load-connected converters, the power/current reference values (p.u.) uniformly span from 0.8 to 1.2.
- For grid-connected, DG-connected, and ESS-connected converters, the power/current reference values (p.u.) are uniformly sampled between 0.9 and 1.1.
- Any single lines/double lines are disconnected at random considering $N-1$ and $N-2$ contingency, respectively. At the same time, the tie switch is closed to maintain an uninterrupted power flow to the loads, which is integral to the subsequent monitoring of converter outages. Furthermore, as part of this process, each terminal is automatically checked during the sampling process to ensure that no independent terminals are present.

A. Network Modelling

In both the conventional CNN and the proposed PECNN, six original input channels are converted to three channels (voltage, current, and conductance), while these three channels are not reformulated in the conventional CNN. The output features are set as 64 in the first CNN encoder block, which are converted to 128, 256, 512 in the following blocks as shown in Fig. 1. In the encoder and residual blocks, the kernel sizes are set to 3×3 and 1×1, respectively. The output is produced using a 1×1 kernel, with both stride s and padding p configured to 1. Rectified linear unit (ReLU) activation function is adopted in

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TABLE I
SYSTEM PARAMETERS OF MVDC NETWORKS.

Converter Parameters		Converter 1	Converter 2	Converter 3	Converter 4	Converter 5	Converter 6	Converter 7	Converter 8	Converter 9	Converter 10
V_{dcrefm} (kV),	Network 1	12, –, –	12, 8, –	12, –, -0.6	12, -8, –	12, –, 0.1	12, –, 0	/	/	/	/
P_{dcrefm} (MW),	Network 2	20, –, –	20, 15, –	20, –, -0.25	20, –, -0.25	20, –, -0.25	20, –, -0.25	20, -5, –	20, -5, –	/	/
I_{dcrefm} (kA)	Network 3	70, –, –	70, –, 0.7	70, 50, –	70, –, -0.65	70, -45, –	70, –, -0.65	70, –, -0.15	70, -10, –	70, –, 0.15	70, –, 0
	Network 1	V	P/V (8.33)	I	P	I/V (0.35)	I/V (0.35)	/	/	/	/
Control Method	Network 2	V	P/V (10)	I/V (0.2)	I/V (0.2)	I	I	P	P	/	/
	Network 3	V	I/V (0.12)	P/V (8.57)	I/V (0.10)	P/V (7.14)	I/V (0.10)	P	I	I/V (0.03)	I/V (0.02)
Line Parameters	**Line 1**	**Line 2**	**Line 3**	**Line 4**	**Line 5**	**Line 6**	**Line 7**	**Line 8**	**Line 9**	**Line 10**	**Line 11**
Distance,	1	4, 0.1, 0.02	5, 0.1, 0.02	5, 0.1, 0.02	4, 0.1, 0.02	3, 0.1, 0.02	/	/	/	/	/
R (Ω/km),	2	6, 0.1, 0.02	8, 0.1, 0.02	8, 0.1, 0.02	6, 0.1, 0.02	4, 0.1, 0.02	4, 0.1, 0.02	4, 0.1, 0.02	/	/	/
L (mH/km)	3	12, 0.1, 0.02	16, 0.1, 0.02	16, 0.1, 0.02	16, 0.1, 0.02	16, 0.1, 0.02	12, 0.1, 0.02	8, 0.1, 0.02	8, 0.1, 0.02	8, 0.1, 0.02	6, 0.1, 0.02

Note 1: The units of I/V and P/V droop constants are kA/kV and MW/kV, respectively.
Note 2: The listed line distance refers to the distance for each pole.

each convolutional layer. Since MLP only accepts the input of one-dimensional data, the inputs of an MLP are n dc voltage, n dc current and $n \times n$ conductance values. The used MLP includes four hidden layers with in total 1000 neurons in all hidden layers.

A total of 10000 examples are generated to train all NNs and 1000 resampled data are used to test the network performance. All NNs are trained for 1000 epochs using the Adam optimizer, with a learning rate of 0.001 and a batch size of 32. The network loss function employs MSE, and the prediction accuracy is determined by the proportion of data deemed qualified to the entire test set. Test data is considered qualified when the error between the sum of actual MVDC voltage values predicted by the NNs and the provided samples is less than a predefined error ϵ, expressed as:

$$error_{sum}^{p.u.} = \left| \sum_{i=1}^{n} V_{dcm,i}^{act(p.u.)} - \sum_{i=1}^{n} V_{dcm,i}^{tar(p.u.)} \right| < \epsilon. \quad (21)$$

B. Case 1: Fixed Network Topology

The topology of all MVDC networks remains consistent, where only the values of converter references, and line resistances are uniformly sampled. The five networks are trained separately and the prediction accuracy is verified in MLP, CNN and proposed PECNN. The comparison results are listed in Table II. The results indicate that the proposed PECNN outperforms other NNs, with its accuracy most notably improving as the number of iterations increases, especially in the 10T-MVDC system.

To further demonstrate the superiority of physics-informed channel reformulation, a contrast experiment is conducted in 10T-MVDC network as shown in Fig. 5. Three NNs are used in the contrast experiment, including MLP, CNN and PECNN (PECNN refers to PECNN with 20 iterations in the subsequent description), to assess their performance based on MVDC voltage prediction errors $error_{sum}^{p.u.}$. Fig. 5(a) and Fig. 5(b) show the prediction accuracy comparison when the dc power/current references of load-connected converters ranges from 0.3 to 1.7 and dc power/current references of grid-connected converters change from 0.7 to 1.3, respectively. The proposed PECNN still has higher prediction accuracy in untrained areas, although all three NNs have good performance in trained areas.

TABLE II
PREDICTION ACCURACY COMPARISON OF DIFFERENT NNS UNDER CASE 1.

Neural Network	6T-MVDC	8T-MVDC	10T-MVDC
MLP	95.8%	91.4%	88.2%
CNN	94.6%	92.2%	87.9%
PECNN (1 iteration)	99.5%	98.0%	97.8%
PECNN (20 iterations)	**99.7%**	**99.0%**	**98.5%**

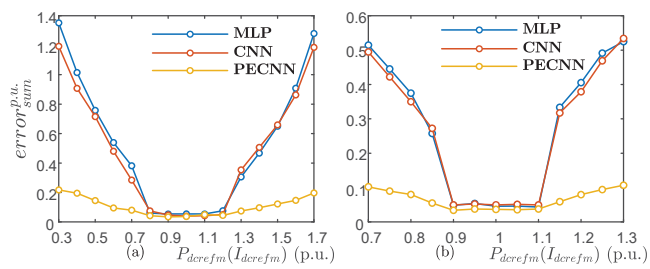

Fig. 5. Prediction error comparison for 10T-MVDC distribution system under different NNs: (a) dc power/current reference value changes of load-connected converters, and (b) dc power/current reference value variations of grid-connected converters.

C. Case 2: Varying Network Topology with Random Line Disconnection

Random line disconnection is included in this case in addition to the uniform sampling of converter references and line resistances. Single and double line disconnections are considered together to mimic $N-1$ and $N-2$ contingencies. Table III lists the prediction accuracy considering random single and double line disconnections in the five MVDC systems. The prediction accuracy of MLP and conventional CNN declines compared to fixed topology due to the increased training complexity for untrained topology structures. However, the proposed PECNN also maintains good performance in such case, despite a slight decrease in prediction accuracy.

In addition, a test for generalization capability is conducted, wherein the system is trained on single line disconnections and then tested on double line disconnections ($N-2$ contingency). Fig. 6 presents a detailed comparison of prediction errors, examining the effects of single-line disconnection (line 1) and concurrent disconnections of line 1 & line 3, within the 10T-MVDC system. The proposed PECNN outperforms others in

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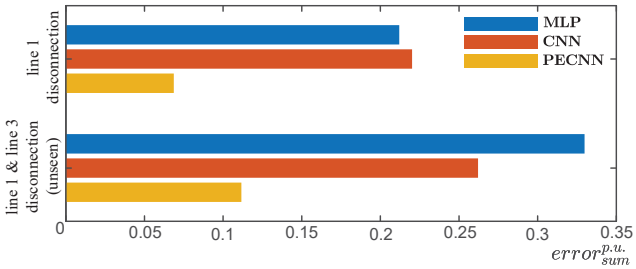

Fig. 6. Prediction error comparison for random line disconnection occurred in 10T-MVDC distribution system under different NNs.

both single and unseen double line tripping scenarios, owing to its integration of physics-based information.

TABLE III
PREDICTION ACCURACY (%) COMPARISON OF DIFFERENT NNS UNDER CASE 2 - LINE DISCONNECTION.

Neural Network	6T-MVDC	8T-MVDC	10T-MVDC
MLP	86.8%	82.8%	78.6%
CNN	90.8%	88.3%	83.5%
PECNN (20 iterations)	**99.4%**	**98.5%**	**98.1%**

V. CONCLUSION

This paper proposes a novel PECNN with integrated physics guidance to predict terminal dc voltage in MT-MVDC distribution networks, addressing the accuracy reduction issue in conventional NNs under unseen data distributions and topology structures. The PECNN simplifies its structure by merging various MVDC bus-type inputs into three channels using a channel combination layer. This reduces the NN computation burden and strengthens the feature extraction capability of PECNN by information concentration across different channels. In the presented physics-informed channel reformulation method, the intrinsic physical relation between voltage, current, and conductance in an MVDC network is embedded by reconstructing the input voltage and current matrices. It allows a more thorough extraction of information, elevates prediction accuracy and strengthens generalization capability.

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